



BIT1123 OBJECT ORIENTED PROGRAMMING

Assignment Title: Object Oriented Programming Fundamentals in Java

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Table of Contents

Introduction.....	3
Knowledge Gained from Each Tutorial	4-5
Challenges Encountered	6
How the Challenges were Overcome	7
Improvements in Java Programming Skills	8
Understanding of Object-Oriented Programming Concepts	9
Future Learning Plans	10
Conclusion	11
GitHub Repository URL	11

1. Introduction

Object-Oriented Programming (OOP) is a well-known programming concept that involves the use of classes and objects to group code into object entities. Throughout the entire BIT1123 Object Oriented Programming course I successfully undertook practical Java tutorials that I relied upon to gain and improve my basic Java programming and OOP fundamentals.

The topics and areas included, but were not limited to classes and objects, encapsulation, inheritance, polymorphism, abstraction, Java ArrayList and file handling, as well as Java Swing GUI development, and experience in utilising the GitHub and GitHub Codespaces tools to organise, store and manage my source code. This report is a reflection of the experience gained, the problems that were encountered throughout the practical tutorials, and the increase in understanding of Java and Object Oriented Programming.

2. Knowledge Gained from Each Tutorial

Week 1 – Introduction to Java and GitHub

Week 1, was about setting up a GitHub repository for my Java tutorial project and organizing my files. I practiced some fundamentals programming in java by writing java program like HelloWorld.java and Student.java. In doing so I learned about the syntax and running java programs in java environment.

Week 2 – Classes and Objects

In Week 2 of the OOP module, I was exposed to the process of creating a class including attributes, constructor, and methods in object oriented programming. Working on the Student class made it quite clear for me how to implement classes and how we use them to create objects in memory and using these objects we are able to access any of the methods of that object.

Week 3–4 – Inheritance and Polymorphism

Week 3 and 4 in this module I studied inheritance with the Person, Student, and Lecturer classes, how child classes are allowed to inheritance from their parent class using extend keywords and then how to override methods and produce different behaviours.

Week 5 – Encapsulation

Week 5: This week I discovered encapsulation. I've used private variables for protecting my student info and implemented getters and setters for modifying and retrieving the student data. This explains why we can't just go directly into data at all times.

Week 6 – Inheritance and the Super Keyword

During Week 6, I practiced inheritance more by creating a Student class and a Staff class and then making Lecturer subclass of Staff. From this I further practice using the super keyword and discovered the different way it could be utilized to call to parent and child class in order to share attributes and methods.

Week 7 – Abstraction

In Week 7, abstraction was addressed. I wrote an abstract class called Appliance and then wrote child appliance classes that inherited from it and implemented the abstract method 'operate'. This way, I had a parent class specifying the general behaviour and child classes implementation for it's operation.

Week 8–9 – ArrayList and File Handling

During this Week 8 and 9, I have been able to learn the functionality of an ArrayList to keep various tasks on. Also, I have had an introduction in using Jave and it's file reading or writing properties by writing the file into a text document and also reading it back into the program.

Week 10 – Java Swing GUI

This 10th week, I have practiced the basis to create a graphical user interface using the Swing and components like JFrame, JLabel, JButton and how use the ActionListener to action a button click in a little quiz.

3. Challenges Encountered

One challenge, I had throughout the tutorials was to grasp how all the Object-Oriented Programming concepts tie together in a real Java code. At the beginning, it was hard to distinguish inheritance, polymorphism, encapsulation, and abstraction due to the fact that all of these principles have the same goal but work for different purposes. As well, I had issues organizing and running my Java files with GitHub and GitHub Codespaces because I needed to ensure I was in the correct folder to be able to run it.

Additionally, I learned during the file handling tutorial that after creating a new file such as a task.txt, I needed to commit and push it for it to show up in the GitHub repository.

It became a bit difficult when learning about multiple Java classes that work with one another in a single program. The programs became more complex as we proceed further in the tutorials, by adding parent classes, child classes, constructors, override methods and etc. And it needed much more focus when compared to basic programs in the earlier tutorials.

4. How the Challenges were Overcome

For this problem, I primarily did practice by walking through tutorials line-by-line, and actually implemented the OOP definitions in Java programs rather than just learning how they work. This helps me understand how inheritance, encapsulation, polymorphism, and abstraction relate to each other. Similarly, on a problem I was struggling with the functionality of Github and Codespaces, I now understand it can be as simple as making sure I'm in the right folder which I learned by checking my folder structure, also with the terminal, know what terminal codes to run before I run my code, and more accurately commit and push files.

When running programs that contain multiple different files, I rechecked how these classes are connected and test programs only once they are finished coding.

By walking through these tutorials repeatedly, the organization of java programs starts to become more-clear.

5. Improvements in Java Programming Skills

Along the way, my Java skills evolved from just writing basic programs to designing programs with multiple classes and many concepts related to Object-Oriented Programming. Confidence and expertise grew in constructing classes, objects, constructors, methods, and variables. I also grew my skills when it comes to methods overriding, inheritance, getters, and setters, abstract classes, and even basic file handling and using ArrayList.

Towards the end of the tutorials, I even acquired some experience with Java Swing and how it can be utilized to implement GUI and event handling in my Java applications.

Apart from the programming and OOP aspect, I learned more about code organisation through GitHub and GitHub codespace.

6. Understanding of Object-Oriented Programming Concepts

From the tutorials, I gained an awareness of the key principles of OOP. These principles involve encapsulation, where data is protected by setting the variables as private and accessing and setting these variables through getter and setter methods. Through Inheritance child class objects are given child object characteristics and class attributes of a parent class to avoid the rewriting of code, which results in less redundant code.

I have also learned the definition of polymorphism which relates to how a parent object represents different types of child objects, also with method overriding a parent may have it characteristics or code overridden in its child objects to gain distinct code. Through Abstraction, characteristics or behaviour can be listed or derived through abstract class definition, leaving the definition of the specific behaviour to the child class.

Through the tutorial application programs, I've found how the principals can be implemented for organising programs within the Java application into class definitions, this way it appears to be that programs based on object oriented programming are far more understandable and reusable.

7. Future Learning Plans

As my next step, I aim to make use of my current java and OOP knowledge by building more practical applications. I hope to expand my knowledge of advanced java, and gain more confidence on designing projects without supervision.

Furthermore, I would like to practice the design of building up bigger OOP applications encompassing multiple concepts including encapsulation, inheritance, polymorphism & abstraction, and even extend on OOP knowledge with java GUI programming, file handling, java I/O and the use of GitHub to create more complete software projects.

Through continuous practice on more complicated tasks and projects, I believe that my ability in problem solving and coding skills will eventually mature.

8. Conclusion

To conclude, through these tutorials the understanding of programming in Java and the basic principles of OOP have been established in my mind. Gradually developing, I learned from simple java programs to classes, inheritance, encapsulation, polymorphism, abstraction, file handling, and GUI, gaining knowledge of java in various levels. I also learned to effectively manage my programs using GitHub in terms of an organized manner which helped me achieve my tasks in order.

Though, having difficulties in the above-mentioned stages; solution to all those problems brought me further confidently to learn more about Java programming. In addition, the knowledge gained here will prove to be an important foundation for further studies related to programming and software development in future.

GitHub Repository URL: https://github.com/vignarohan/Shri_object-oriented-programming-tutorials